WORKED SOLUTION

STEP III 2017, Question 8

Pure • Summation by Parts

STEP III 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q8

The key idea

The opening identity is the discrete version of integration by parts ("summation by parts"): it trades a sum involving differences of b for one involving differences of a. To use it on a trigonometric sum, we pick b_m so that $b_{m+1} - b_m$ produces the terms we want; the sum-to-product formula turns a difference of sines or cosines into a single product. Choosing a_m to be 1 or m then makes the other side easy.

Solution

The identity

The quantity $a_{n+1}b_{n+1} - a_1b_1$ telescopes:

$$a_{n+1}b_{n+1} - a_1b_1 = \sum_{m=1}^{n} (a_{m+1}b_{m+1} - a_m b_m).$$

Adding and subtracting $a_m b_{m+1}$ inside each bracket,

$$a_{m+1}b_{m+1} - a_m b_m = a_m(b_{m+1} - b_m) + b_{m+1}(a_{m+1} - a_m).$$

Summing over $m = 1, \ldots, n$ and rearranging gives exactly the required identity.

Throughout the trigonometric parts we assume $\sin \frac{1}{2}x \neq 0$ (that is, x is not a multiple of 2π), so that $\operatorname{cosec} \frac{1}{2}x$ is defined.

Part (i)

With $b_m = \sin mx$, the sum-to-product formula with $A = (m+1)x$, $B = mx$ gives

$$b_{m+1} - b_m = \sin(m+1)x - \sin mx = 2\cos(m + \tfrac{1}{2})x \, \sin \tfrac{1}{2}x.$$

Now take $a_m = 1$ for every m, so that $a_{m+1} - a_m = 0$ and the final sum in the identity vanishes. The identity becomes

$$2\sin\tfrac{1}{2}x \sum_{m=1}^{n} \cos(m+\tfrac{1}{2})x = b_{n+1} - b_1 = \sin(n+1)x - \sin x.$$

Dividing by $2\sin\frac{1}{2}x \neq 0$,

$$\sum_{m=1}^{n} \cos(m+\tfrac{1}{2})x = \tfrac{1}{2}(\sin(n+1)x - \sin x)\operatorname{cosec}\tfrac{1}{2}x,$$

as required.

Part (ii)

The factor m suggests $a_m = m$, so that $a_{m+1} - a_m = 1$. We then need $b_{m+1} - b_m$ to be a multiple of $\sin mx$.

Ask yourself Which choice of b_m makes $b_{m+1} - b_m$ a multiple of $\sin mx$?

In part (i), the difference of two sines at $(m+1)x$ and mx gave a cosine at the *midpoint* $(m+\frac{1}{2})x$. To get $\sin mx$ we want a difference of cosines whose arguments have midpoint mx, so we take $b_m = \cos(m-\frac{1}{2})x$. Using $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ (which also follows from the first note in the question),

$$b_{m+1} - b_m = \cos(m+\tfrac{1}{2})x - \cos(m-\tfrac{1}{2})x = -2\sin mx\,\sin\tfrac{1}{2}x.$$

Write $S = \sum_{m=1}^{n} m\sin mx$. With $a_m = m$ the identity gives

$$-2\sin\tfrac{1}{2}x\, S = (n+1)\cos(n+\tfrac{1}{2})x - \cos\tfrac{1}{2}x - \sum_{m=1}^{n}\cos(m+\tfrac{1}{2})x.$$

The last sum is exactly part (i). Substituting it and multiplying through by $-2\sin\frac{1}{2}x$:

$$4\sin^2\tfrac{1}{2}x\, S = 2\sin\tfrac{1}{2}x\cos\tfrac{1}{2}x - 2(n+1)\cos(n+\tfrac{1}{2})x\,\sin\tfrac{1}{2}x + \sin(n+1)x - \sin x.$$

Now simplify the two products:

- $2\sin\frac{1}{2}x\cos\frac{1}{2}x = \sin x$, which cancels the $-\sin x$;
- by $2\cos A\sin B = \sin(A+B) - \sin(A-B)$ with $A = (n+\frac{1}{2})x$, $B = \frac{1}{2}x$,

$$2\cos\big(n+\tfrac{1}{2}\big)x\,\sin\tfrac{1}{2}x = \sin(n+1)x - \sin nx.$$

Hence

$$\begin{aligned} 4\sin^2\tfrac{1}{2}x\, S &= -(n+1)\big(\sin(n+1)x - \sin nx\big) + \sin(n+1)x \\ &= -n\sin(n+1)x + (n+1)\sin nx. \end{aligned}$$

Dividing by $4\sin^2\frac{1}{2}x$,

$$\sum_{m=1}^{n} m\sin mx = \left(-\tfrac{n}{4}\sin(n+1)x + \tfrac{n+1}{4}\sin nx\right)\operatorname{cosec}^2\tfrac{1}{2}x,$$

so

$$p = -\frac{n}{4}, \qquad q = \frac{n+1}{4}.$$

Checks

- For $n = 1$ the formula gives $\frac{1}{4}(2\sin x - \sin 2x)\operatorname{cosec}^2\frac{1}{2}x = \frac{1}{2}\sin x\,(1-\cos x)\operatorname{cosec}^2\frac{1}{2}x = \sin x$, using $1-\cos x = 2\sin^2\frac{1}{2}x$. That is the single term $1\cdot\sin x$.
- Numerically, with $n = 3$ and $x = 1$: $\sin 1 + 2\sin 2 + 3\sin 3 \approx 3.08343$, and $\left(-\frac{3}{4}\sin 4 + \sin 3\right)\operatorname{cosec}^2\frac{1}{2} \approx 3.08343$ too.
- Part (ii) can also be found by differentiating $\sum_{m=1}^{n}\cos mx$ with respect to x, which gives $-S$; this is a useful independent check.

Common slips

- Choosing $b_m = \cos mx$ in part (ii): the difference is then a multiple of $\sin(m+\frac{1}{2})x$, not $\sin mx$, and the method stalls.
- Sign errors in $\cos A - \cos B$: it is $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$, with a minus sign.
- Forgetting to state that $\sin\frac{1}{2}x \neq 0$ before dividing by it.

Beyond STEP

Summation by parts (Abel summation) is the discrete twin of integration by parts, and it is a workhorse of first-year analysis. Combined with bounds like part (i), which shows the partial sums of $\cos(m+\frac{1}{2})x$ stay bounded, it proves Dirichlet's test: for example, $\sum\frac{\sin mx}{m}$ converges for every x. Sums like part (ii) also appear in Fourier series, where such closed forms are the starting point for studying convergence. Abbott's *Understanding Analysis* treats Abel summation and Dirichlet's test clearly.

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